

11.1 Atoms, Nuclei & Radiation

Question Paper

Course	CIE A Level Physics
Section	11. Particle Physics
Topic	11.1 Atoms, Nuclei & Radiation
Difficulty	Medium

Time allowed: 40

Score: /28

Percentage: /100

Question 1a

A nucleus X decays by emitting a β^+ particle to form a new nucleus, ${}_{11}^{23}\text{Na}$.

State the number of protons and the number of neutrons in nucleus X.

number of protons =

number of neutrons =

[2 marks]

Question 1b

State **one** similarity between a β^+ particle and a β^- particle.

[1 mark]

Question 1c

State the quark composition of a meson.

[1 mark]

Question 1d

A hadron consists of two down quarks and a charm quark.

Determine the charge of the hadron. Show your working.

charge = C

[2 marks]

Question 2a

The deflection of α -particles by a thin metal foil is investigated with the arrangement shown in Fig 1.1.

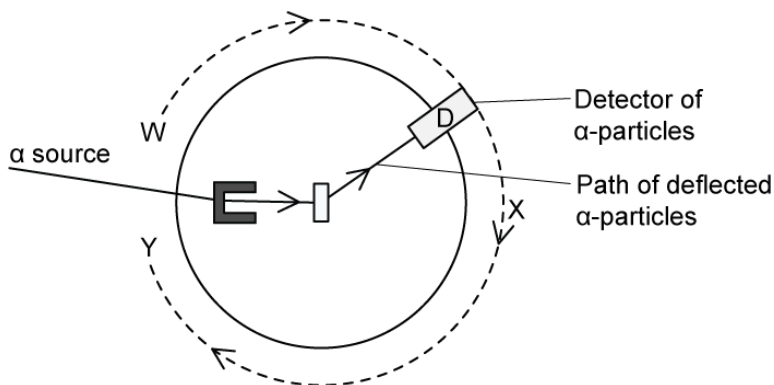


Fig 1.1

The detector of α -particles, D, is moved around the path labelled WXY.

State the environment the apparatus must be enclosed in and explain why this is required for the experiment.

[2 marks]

Question 2b

Explain the readings detected by D and the conclusions that were made about the nature of atoms at points

(i)
W,

[2]

(ii)
X.

[2]

[4 marks]

Question 2c

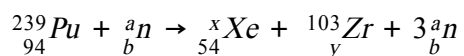
State what the readings detected by D at position Y conclude about the charge of the nucleus. Explain why.

[2 marks]**Question 2d**

A beam of α -particles produces a current of 3.5 pA. Calculate the number of α -particles per second passing a point in the beam.

[3 marks]**Question 3a**

When a nucleus of plutonium-239 absorbs a neutron, the following reaction may take place.



State the values of a , b , x and y .

[3 marks]

Question 3b

State two quantities, other than the proton and neutron number, that are conserved in the fission reaction in part (a).

[2 marks]

Question 3c

A Plutonium-239 nucleus has a mean density of $1.3 \times 10^{17} \text{ kg m}^{-3}$.

Show that the radius of a Plutonium-239 nucleus is $9.0 \times 10^{-15} \text{ m}$.

[3 marks]

Question 3d

Plutonium-239 is an α -emitter whilst Uranium-237 is a β -emitter.

Aside from their structure, distinguish between an α -particle and a β -particle.

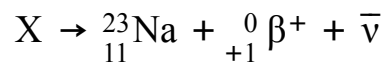
[3 marks]



Model Answers: Medium

1a

(a) For the decay:



Nucleus X has:

- Mass number = $23 + 0 = 23$ protons + neutrons
- Number of protons = $11 + 1 = 12$ protons; [1 mark]
- Number of neutrons = $23 - 12 = 11$ neutrons; [1 mark]

[Total: 2 marks]

1b

(b) β^+ and β^- particles have the same:

Either **one** from:

- (Rest) mass; [1 mark]
- (Magnitude of) charge; [1 mark]

[Total: 1 mark]

1c

(c) A meson is made up of:

- A quark and an anti-quark; [1 mark]

[Total: 1 mark]

1d

(d) Determine the charge of the hadron with composition ddc :

State the charge on each quark:

- Down quark = $-\frac{1}{3}e$

AND

- Charm quark = $+\frac{2}{3}e$ [1 mark]

Calculate the charge on the hadron:

- Charge on $ddc = -\frac{1}{3}e - \frac{1}{3}e + \frac{2}{3}e = 0$ [1 mark]

[Total: 2 marks]

2a

(a) State the environment the apparatus must be enclosed in and explain why this is required for the experiment.

The apparatus must be enclosed in:

- A vacuum; [1 mark]

Because:

- α -particles travel / are absorbed by a short distance in air; [1 mark]

[Total: 2 marks]

2b

(b) Explain the readings detected by D and the conclusions that were made about the nature of atoms at points

(i) W:

- A very small proportion deviated in the backwards direction / at large angles; [1 mark]
- This shows that the mass and charge are concentrated in a (very small) nucleus; [1 mark]

(ii) X:

- The majority went straight through; [1 mark]
- This shows that most of the atom is empty space / the nucleus is very small compared with the atom; [1 mark]

[Total: 4 marks]

2c

(c) State what the readings detected by D at position Y conclude about the charge of the nucleus. Explain why.

- The charge of the nucleus is positive; [1 mark]
- Because the alpha particle is positive **AND** it is repelled / deflected (at an angle); [1 mark]

[Total: 2 marks]

Remember the alpha particle is just a helium **nucleus** with two protons and two neutrons. Only protons have a charge in the nucleus (which is positive) so all nuclei are always positive.

2d

(d) Calculate the number of α -particles per second passing a point in the beam

List the known quantities:

- Current, $I = 3.5 \text{ pA} = 3.5 \times 10^{-12} \text{ A}$

State the equation for current, I :

- $Q = It \Rightarrow I = \frac{Q}{t}$ [1 mark]

Determine an equation for the number of α -particles, n :

- $I = n \times \frac{Q}{t} \Rightarrow \frac{n}{t} = \frac{I}{Q}$

- $\frac{n}{t} = \frac{3.5 \times 10^{-12}}{2 \times (1.60 \times 10^{-19})}$ [1 mark]
- $\frac{n}{t} = 1.094 \times 10^6 = 1.1 \times 10^6 \text{ s}^{-1}$ [1 mark]

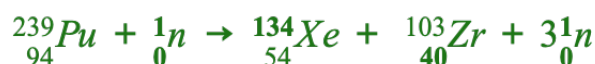
[Total: 3 marks]

An alpha particle is a helium nucleus of **2** protons and **2** neutrons. Therefore, its charge is the $2 \times$ charge of a proton ($e = 1.60 \times 10^{-19} \text{ C}$). You will be expected to remember the composition of an alpha particle, and the charge e is given on your data sheet.

The wording 'per second' means 'every 1 second'. Therefore, the 'number of particles, n , per second' is represented as an equation $\frac{n}{t}$.

3a

(a) The values of a , b , x and y are:



- $a = 1$ **AND** $b = 0$ [1 mark]

Calculate x :

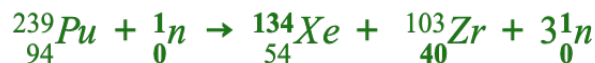
- $239 + 1 = x + 103 + (3 \times 1)$
- $x = 240 - 106$
- $x = 134$ [1 mark]

Calculate y :

- $94 + 0 = 54 + y + (3 \times 0)$
- $y = 94 - 54$
- $y = 40$ [1 mark]

[Total: 3 marks]

On the right-hand side, $3{}_0^1n$ means that there are **3** neutrons, so **both** the mass number (1) and the atomic number (0) must be **multiplied** by 3.



- $a = 1$ **AND** $b = 0$ [1 mark]

Calculate x :

- $239 + 1 = x + 103 + (3 \times 1)$
- $x = 240 - 106$
- $x = 134$ [1 mark]

Calculate y :

- $94 + 0 = 54 + y + (3 \times 0)$
- $y = 94 - 54$
- $y = 40$ [1 mark]

[Total: 3 marks]

On the right-hand side, $3{}_0^1\text{n}$ means that there are **3** neutrons, so **both** the mass number (1) and the atomic number (0) must be **multiplied** by 3.

3b

(b) Three quantities that are conserved in the fission reaction in part (a) are:

- mass and energy / mass-energy; [1 mark]
- momentum; [1 mark]

[Total: 2 marks]

Be specific in your wording. Just 'mass' or 'energy' conservation would not be accepted here, since energy and mass are interchangeable and are often referred to as the same thing in particle physics (e.g. rest-mass energy).

3c

(c) Calculate the radius of a Plutonium-239 nucleus

List the known quantities:

- Mean density of the Plutonium-239 nucleus, $\rho = 1.3 \times 10^{17} \text{ kg m}^{-3}$

Calculate the mass of the Plutonium nucleus, m :

- $m = \text{mass number} \times \text{atomic mass unit} = 239 \times 1 \text{ u}$
- $m = 239 \times (1.66 \times 10^{-27}) = 3.9674 \times 10^{-25} \text{ kg}$ [1 mark]

Calculate the radius of the Plutonium nucleus, r .

- $\rho = \frac{m}{V} = \frac{m}{\frac{4}{3}\pi r^3} = \frac{3m}{4\pi r^3}$
- $r = \sqrt[3]{\frac{3m}{4\pi\rho}} = \sqrt[3]{\frac{3 \times (3.9674 \times 10^{-25})}{4\pi \times (1.3 \times 10^{17})}}$ [1 mark]
- $r = 8.996 \times 10^{-15} = 9.0 \times 10^{-15} \text{ m}$ [1 mark]

[Total: 3 marks]

You can approximate a nucleus to a sphere, but you must remember the equation for the volume of a sphere, as this will not be given on your data sheet! However, the value of the atomic mass unit (u) will.

3d

(d) Aside from their structure, distinguish between an α -particle and a β -particle.

Any **three** from:

- The speed of an α -particle is less than the speed of a β -particle; [1 mark]
- An α -particle has discrete values of energy / speed whilst β -particles have a continuous spectrum / range of energies; [1 mark]
- α -particles have greater ionising power than β -particles **OR** β -particles have a shorter range than α -particles; [1 mark]
- α -particles are positive whilst β -particles are negative; [1 mark]

[Total: 3 marks]

Reverse arguments would be accepted for all of these answers e.g. the speed of a β -particle is greater than the speed of an α -particle. The main thing is to remember that a comparison must mention both.

Read the question carefully, as it says 'aside from their structure'. This means answers such as 'an α -particle is two protons and two neutrons and a β -particle is an electron' will not be accepted.